



Semantic-Aware Developer Influence Analysis using Social Network Analysis and LDA Topic Modeling

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Introduction

- Modern software repositories such as GitHub contain large volumes of developer interaction data through issue discussions, pull requests, and collaborative communication. Traditional Social Network Analysis (SNA) techniques identify influential developers using structural metrics such as Degree Centrality and PageRank. However, these approaches treat all interactions equally regardless of semantic importance.
- This project proposes a semantic-aware developer influence analysis framework that combines Social Network Analysis, lightweight semantic classification, and LDA topic modeling to improve influence measurement in open-source software repositories.

Problem & Research Objectives

- Existing MSR approaches measure developer influence mainly through interaction frequency and graph structure. Simple comments such as “thanks” are treated similarly to important technical discussions such as bug fixing or design improvement. This may produce inaccurate influence rankings.
- Therefore, there is a need for a lightweight semantic-aware influence model that considers both interaction structure and semantic importance.

Research Objectives

- Construct a developer interaction network from GitHub repositories
- Apply Social Network Analysis techniques
- Perform LDA topic modeling on developer discussions
- Classify interactions into semantic categories
- Assign semantic weights to interactions
- Compute semantic-aware developer influence scores
- Compare baseline SNA with semantic-aware SNA

Expected Outcome & Significance

- Results demonstrated that semantic weighting significantly changed developer influence rankings.

Key Findings:

- Technical discussions contributed higher influence scores
- Social comments received lower semantic weights
- LDA successfully identified dominant repository discussion topics.
- Semantic-aware PageRank produced more meaningful developer rankings

Dataset & Tools

Repository Used:

- microsoft/vscode* (GitHub)

Collected Data:

- Issue discussions
- Issue comments
- Developer usernames
- Timestamps

Tools & Technologies:

- Python
- NetworkX
- Pandas
- Gensim
- NLTK
- Matplotlib
- Gephi

Proposed Research Methodology

- GitHub Data Collection
- Text Preprocessing
- LDA Topic Modeling
- Topic Discovery
- Developer Interaction Network Construction
- Baseline Social Network Analysis
- Semantic Classification
- Semantic Weight Assignment
- Weighted Network Construction
- Semantic-Aware Influence Analysis
- Comparison & Visualization

The complete methodology and workflow is shown in the Figure 1.

Conclusion

The proposed approach improved traditional influence analysis by incorporating semantic importance into developer interactions. Experimental results showed that semantic weighting and topic modeling provide more meaningful influence rankings within software repositories.

Future Enhancements

- Apply advanced NLP techniques
- Use transformer-based language models
- Analyze pull request discussions
- Perform topic-aware influence analysis
- Extend to multiple repositories

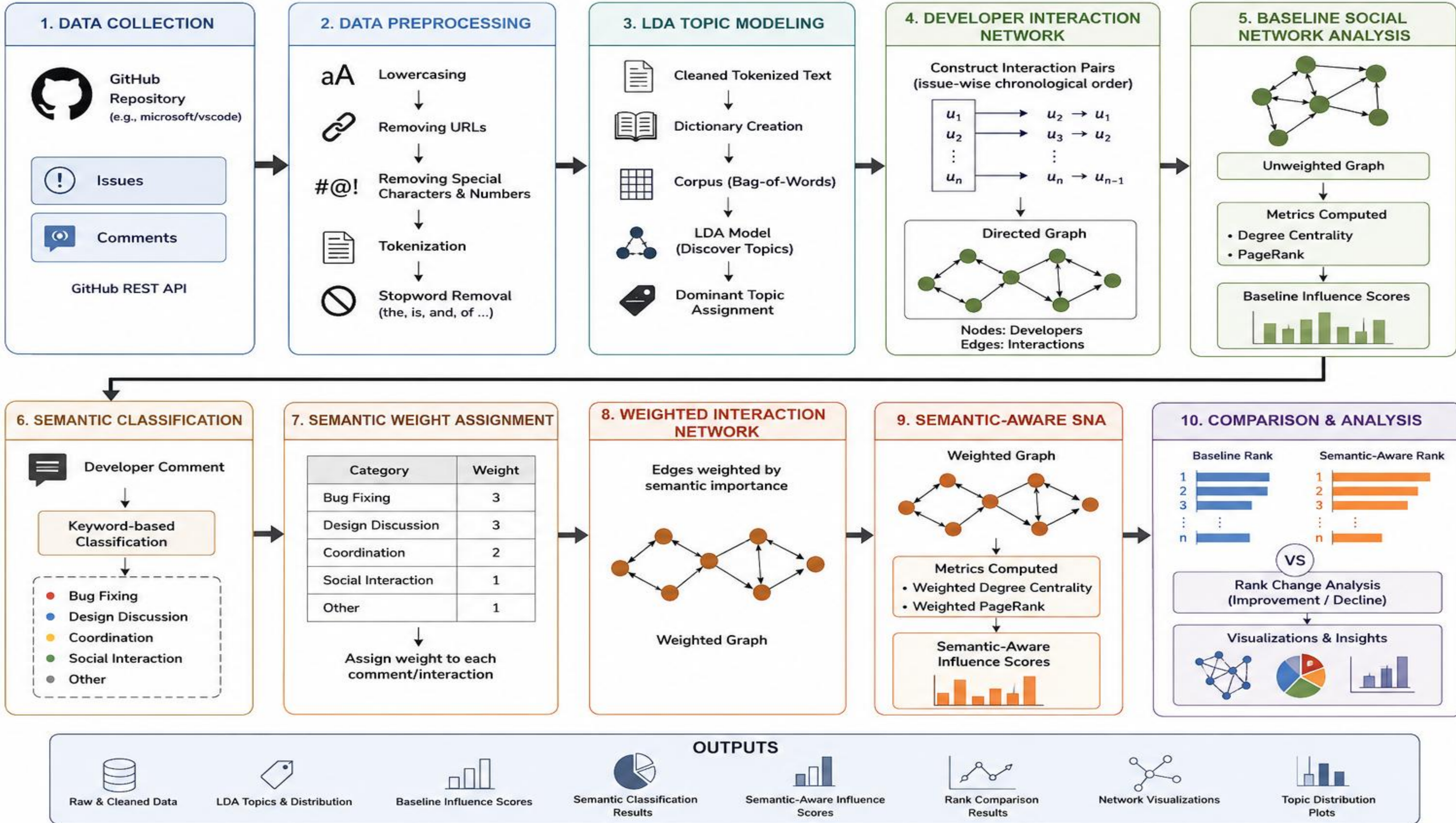


Figure 1. The proposed framework and methodology

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Key References:

- GitHub REST API Documentation
- NetworkX Documentation
- Gensim LDA Documentation
- Gephi Network Analysis Tool